

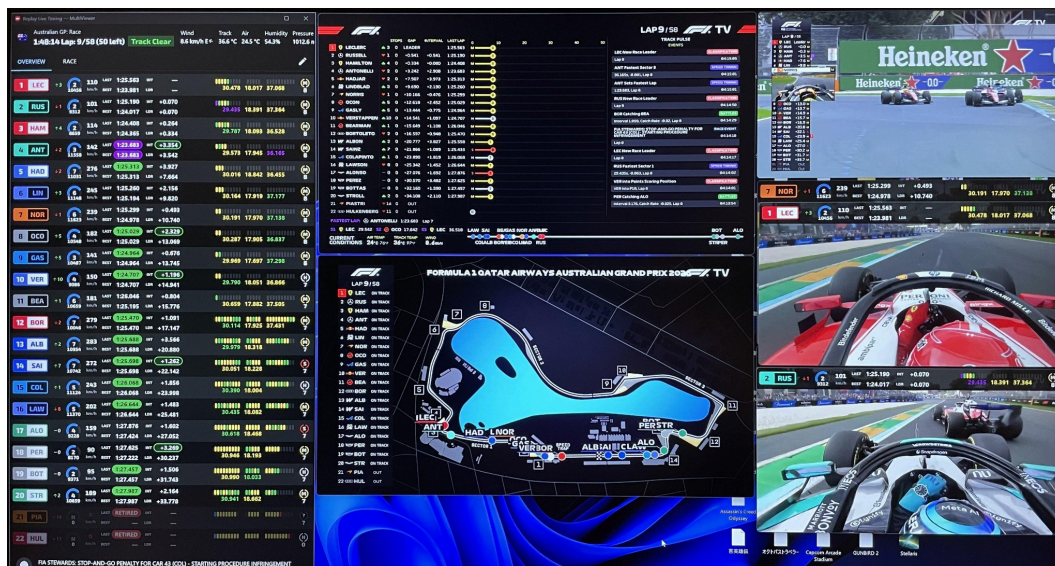
Existing F1 Interfaces: A Comparative Analysis

Objective

This document examines four common ways of following a Formula 1 race: a timing table, a driver tracker, a tyre-history display, and a third-party race dashboard. The goal is to describe what each interface communicates and where a viewer who is still learning Formula 1 may get lost. The screenshots are annotated first, then compared in more detail.

Most of the information is useful but the difficulty is that they often assume the viewer already knows the vocabulary, the color system, and the race situation. An experienced fan can look at a gap, a tyre symbol, or a cluster of cars and understand the story but a novice may see the same display and still not know what changed or why it matters.

Examples of F1 Interfaces



Driver	Gap	Best lap	Last lap	Sector 1	Sector 2	Sector 3	S1 μ S	S2 μ S	S3 μ S	Tyre history
1 VER	—	1:27.625	1:27.973	17.584	38.400					18 (H) 23 (M) 0
2 PIA	+24.457	1:28.927	1:32.723	38.028						0 (M) 41 (H) 0
3 NOR	+29.061	1:27.814	1:48.266	17.929						1 (H) 24 (H) 16 (M)
4 LEC	+35.015	1:28.044	1:28.125	17.856						5 (M) 23 (H) 16 (M)
5 RUS	+35.071	1:28.648	1:29.300	17.768						27 (H) 14 (M) 0
6 STR	+50.339	1:29.719	1:30.029	18.062						42 (H) 0 0
7 ALO	+53.434	1:29.420	1:29.527	17.769						26 (H) 17 (M) 0
8 OCO	+54.180	1:28.740	1:29.435	17.654						22 (H) 19 (M) 0
9 BEA	+55.830	1:29.263	1:29.816	17.845						27 (H) 14 (M) 0
10 BOR	+56.817	1:29.708	1:31.051	18.065						26 (H) 15 (M) 0
11 HUL	+57.697	1:28.442	1:30.223	18.082	38.590	33.551				34 (H) 7 (S) 0
12 HAM	+58.135	1:28.094	1:28.693	17.898	37.927	32.868				10 (M) 23 (H) 8 (S)
13 SAI	+59.521	1:29.212	1:30.502	18.076	38.724	33.702				23 (H) 18 (M) 0
14 HAD	+62.622	1:29.381	1:29.381	17.736	38.777	32.868				27 (H) 14 (M) 0
15 TSU	+64.116	1:28.572	1:28.857	17.670	38.210	32.977				9 (M) 32 (H) 0
16 ANT	+69.102	1:28.029	1:28.130	17.729	37.820	32.581				9 (M) 32 (H) 0
17 LAW	+70.327	1:28.886	1:30.032	17.739	38.831	33.462				20 (H) 21 (M) 0
18 ALB	+71.850	1:28.411	1:28.777	17.783	38.126	32.868				8 (M) 25 (H) 8 (S)
19 GAS	1 L	1:29.442	1:31.907	38.635	38.690					3 (M) 27 (H) 13 (M)
20 COL	1 L	1:29.430	1:33.034	39.369	38.412					0 (M) 25 (H) 15 (M)

Readings

1. Malik's Platform 3 Solutions article: This article discusses data overload, ineffective color use, and the limits of one-size-fits-all dashboards.
2. Bowers's Toptal article : makes a similar point from a design perspective. Explaining how a visualization can be accurate and still fail if it includes too much information, uses color without enough support, or makes the viewer work too hard to identify the important pattern.

Interfaces reviewed

- Official F1 live timing table
- F1 TV driver tracker
- Official F1 tyre-history view
- Graphprix Pitwall third-party dashboard

1. Timing tables

A timing table is the most information-dense of the official race displays. It shows the running order and lets the viewer compare all twenty drivers at once. Depending on the mode, it may also show gaps, lap times, sector times, tyre information, pit status, and team colors. For an experienced viewer, this is often the quickest way to understand the whole field. For a novice, it can look like a wall of abbreviations and constantly changing decimals.

		BEST LP	30	199	342	43.672	216	28.293	236	1:42.164		
1	5 VET	1:42.661			310	49.430	199	29.984	217	15:07.097	13S	7
2	7 RAI	1:42.724	31	299	310					1:42.706	15S	6
3	44 HAM	1:42.798			258	52.157	189	35.835		IN PIT	10S	5
4	77 BOT	1:43.464			275	51.614	192	34.774		IN PIT	13S	8
5	33 VER	1:44.399	30	436	332	45.125	211			24:04.026	9S	2
6	3 RIC	1:44.479	30	573	339	46.265	209			20:24.958	12S	2
7	16 LEC	1:45.159	30	874	318	44.886	211			20:25.333	11S	1
8	11 PER	1:45.341	30	697	337					23:02.589	10S	2
9	27 HUL	1:45.464					211			17:08.111	16S	1
10	31 OCO	1:45.485	30	407	337					9:34.520	13S	2
11	8 GRO	1:45.814	32	453	333					10:30.858	12S	3
12	9 ERI	1:45.865	30	930	323	45.498	208			24:52.848	10S	2
13	55 SAI	1:45.925	30	518	332	49.877	196			19:20.041	13S	2
14	20 MAG	1:46.087	31	985	288					10:01.194	12S	3
15	10 GAS	1:46.182	31	213	323					22:06.047	16S	1
16	28 HAR	1:46.259	30	670	342					23:20.613	17S	2
17	18 STR	1:46.502			301	53.593	198	30.996	225	26:21.809	10S	2
18	35 SIR	1:46.630			298	50.475	199	31.407	218	11:03.031	11S	3
19	14 ALO	1:46.942	31	913	319					11:05.873	16S	3
20	2 VAN	1:47.061			268					IN PIT	14S	1

Official F1 timing table with three novice-viewer issues marked in red.

Source image: Motorsport.com, F1 live timing article.

1. The gap needs a reference point. The display may switch between ‘Interval’ and ‘To Leader’. ‘To Leader’ measures each driver's gap from the race leader. Interval measures the gap from the driver immediately ahead. The same driver can therefore have two different-looking numbers depending on the selected mode. The label does not explain the difference to someone seeing it for the first time.

2. Several numbers change at once. A lap can update positions, gaps, lap times, and sector colors within seconds. The table is technically current, but the motion makes it difficult to decide where to look. A new viewer may follow one driver and miss a lead change or pit stop elsewhere in the table.

3. Abbreviations save space but require prior knowledge. Driver codes such as HAM, VER, and LEC are efficient once they are familiar. The same is true of sector labels, pit indicators, and compound letters. But before that knowledge is learned, the viewer has to translate the code before interpreting the race information.

What the timing table communicates well

The strongest feature of the timing table is coverage. It keeps the full field visible rather than focusing only on the cars shown by the broadcast camera. It is useful for answering specific questions: Who is leading? How far apart are two drivers? Who is gaining? Who has just set a fast sector? It also makes small changes measurable. A track map can show that two cars are close, but the timing table can show whether the gap is 0.4 seconds or 1.2 seconds.

Where a novice may struggle

The main problem is that the rows, columns, color changes, and abbreviations all demand attention at the same time. This matches the readings' warning about data overload. The display gives the viewer almost everything, but it does not clearly say which change is most important at that moment. A novice must create that hierarchy alone.

Color can also be unclear without a legend. Purple, green, and yellow sector times have specific meanings in F1, but those meanings have to be learned. Bowers argues that color should not carry meaning by itself. The timing table partly supports color with numbers, yet the viewer still needs outside knowledge to understand why a purple sector is more significant than a green one.

2. Driver trackers

The driver tracker replaces rows of numbers with a map of the circuit. Each car appears as a colored marker or driver abbreviation positioned around the track. This is usually easier to understand at first glance because it answers a visual question: where are the cars? It can reveal groups, clean air, traffic, and the approximate distance between drivers without requiring the viewer to read a decimal gap.



F1 TV driver tracker

Source image: Formula1.com, F1 TV feature page.

- 1. The map gives an immediate picture of the field.** The viewer can see which cars are isolated and which cars are running in groups. This is more intuitive than comparing twenty separate timing gaps. The shape of the circuit also helps connect the data display to the race footage.
- 2. The side list repeats information but still needs context.** The position list helps identify the order, yet the viewer must look back and forth between the list and the map. A marker can also appear physically close to another car even when the cars are on different laps or separated by a pit stop cycle. Track position is not always race position.
- 3. Car labels become crowded in battles.** When several cars are close on track, their markers and abbreviations overlap. The map shows that a battle exists but can become less readable as the battle gets tighter.

What the tracker communicates well

The tracker is great at spatial relationships. It shows the field as a moving shape rather than a list. A viewer can quickly notice a train of cars, a large empty section of track, or a driver approaching traffic. It is also useful during pit stops because the viewer can watch where a car returns relative to nearby cars.

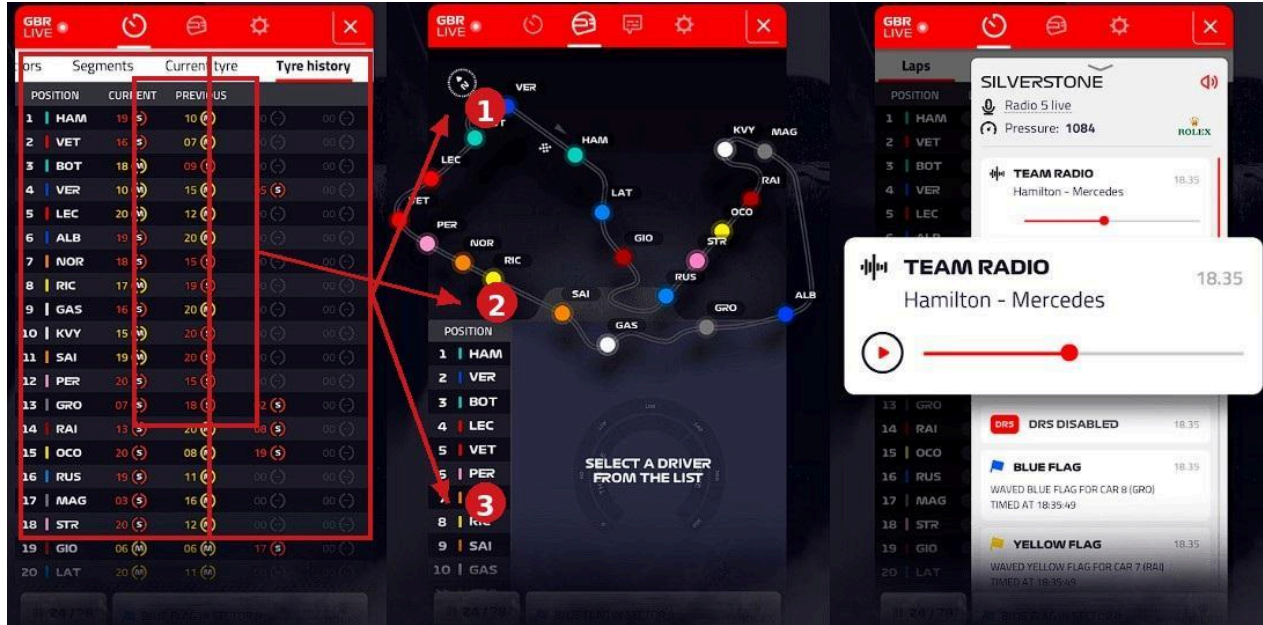
Where a novice may struggle

The map is simpler, but it leaves out important explanations. It does not automatically tell the viewer whether two nearby cars are racing each other for position, whether one is a lap behind, or whether one still needs to make a pit stop. A novice may correctly see proximity but draw the wrong conclusion from it. The map therefore reduces numerical load while increasing the need for race context.

The clutter problem also supports Malik's criticism of one-size-fits-all dashboards. The same marker size that works when cars are spread out becomes difficult when the field bunches up behind a safety car or during the opening lap. The interface does not adapt enough to the situation that creates the most overlap.

3. Tyre displays

Tyre information is important because tyre choice affects pace, grip, pit-stop timing, and race strategy. The official display uses compound letters and colors to show whether a driver is on the soft, medium, or hard tyre. It may also show previous stints and a number representing how many laps have been completed on the current set.



Official F1 tyre-history view with compound and tyre-age issues marked in red.

Source image: F1.com.es, F1 TV interface article.

- 1. The compound uses two cues.** The letters S, M, and H stand for Soft, Medium, and Hard. Color provides a second cue: red, yellow, and white. Using both a letter and a color is helpful because the information does not depend entirely on color perception. However, neither cue explains what the compounds mean for pace, durability, or strategy.
- 2. Tyre age is present but easy to misread.** The number next to the compound shows the number of laps driven on that set, but the unit laps is not always written. A viewer who does not know the convention may read it as a time, a position, or another status value. A small label would remove that ambiguity.
- 3. Tyre history records what happened without explaining why it matters.** Previous compounds show the driver's stint sequence, which is useful for reconstructing strategy. The display does not say whether a stop was early, whether the driver is likely to stop again, or how the current tyres compare with a nearby rival's tyres.

What the tyre display communicates well

The display is compact and supports quick comparison once the compound system is learned. A viewer can identify different strategies across the field and see whether a driver has changed compounds. The use of letters alongside colors follows Bowers's recommendation not to rely on color alone.

Where a novice may struggle

The display presents tyre data assuming the meaning of the data is obvious. Knowing that one driver has ten-lap-old mediums and another has two-lap-old softs does not automatically tell a beginner which driver has the advantage. Soft tyres are usually faster but wear sooner; hard tyres usually last longer but may be

slower or harder to warm. Without that basic explanation, the screen shows a strategic difference without helping the viewer interpret it.

The colors can also create false confidence. Red may look like a warning and white may look neutral, but in this context they are simply compound identities. Malik's point about ineffective color use applies here: a color system works only when its meaning is clear to the audience.

4. Third party dashboards

Third-party dashboards are designed for viewers who want more control than the standard broadcast provides. Graphprix Pitwall combines classification, telemetry traces, track position, battles, sector times, pit stops, alerts, and tyre risk on one screen. It is powerful, but it also makes the difference between expert and novice use especially visible.

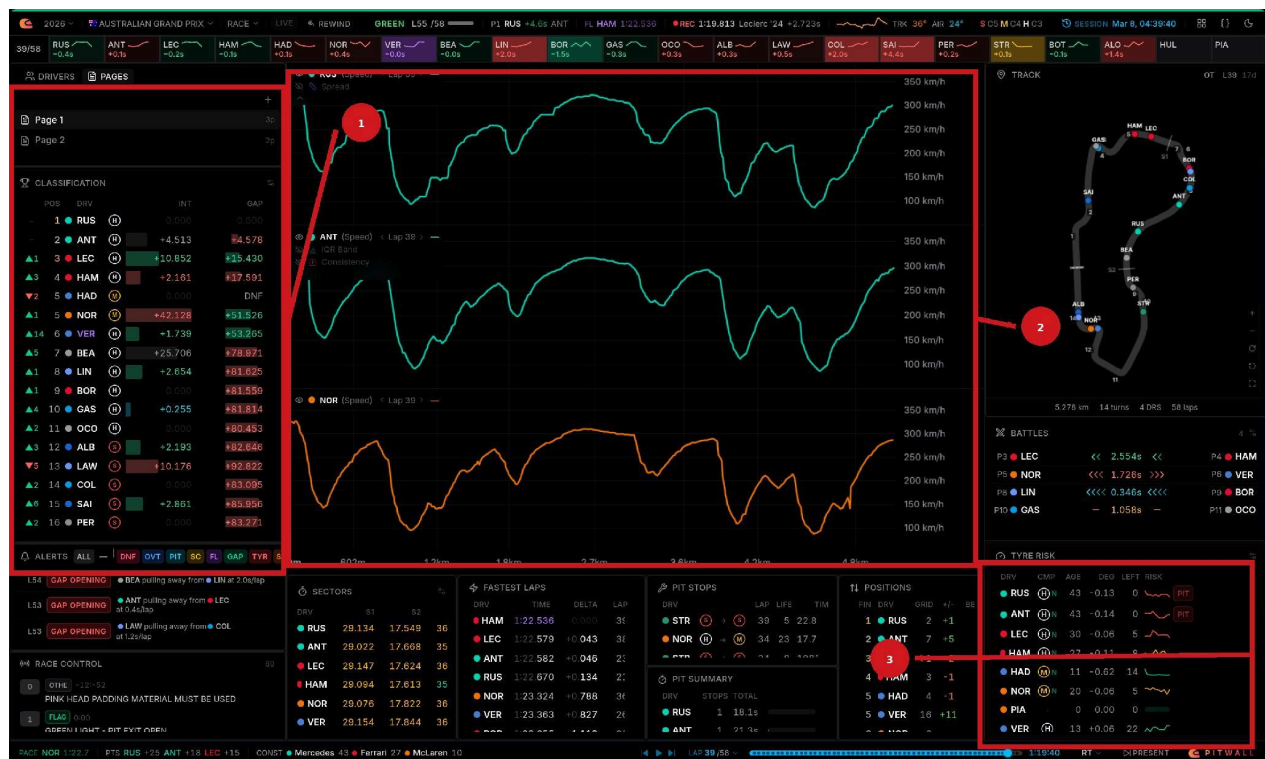


Figure 4. Graphprix Pitwall dashboard, information-density and labeling issues marked in red.

Source image: Graphprix Pitwall.

- 1. Several tasks share one screen.** The classification panel, race alerts, telemetry, map, battle list, pit-stop information, and tyre information all compete for attention. The dashboard is useful when the viewer has a specific question. Without a question, it is difficult to know which panel should be read first.
- 2. Telemetry supports deep comparison.** The speed traces can show where one driver gains time on another. This is information the official broadcast often summarizes rather than displaying directly. A novice, however, must understand the axes, the selected lap, and why the lines rise or fall before the graph becomes useful.
- 3. A small label makes tyre age clearer.** The tyre-risk panel includes an AGE heading which makes the number easier to interpret than the unlabeled tyre-age number in the official display. It is an example of how direct wording can reduce confusion without removing the detail.

What the dashboard communicates well

The dashboard is strongest when the viewer wants to investigate a specific event. It can compare two drivers' pace, check the position of a car after a pit stop, or follow a battle that the broadcast is not showing. It also keeps several kinds of evidence in one place, so an experienced viewer can connect timing, position, and telemetry.

Where a novice may struggle

The screen does not provide a clear starting point. Every panel appears important, and several panels update at once. This is the clearest example of the data-overload problem described in both readings. Nothing is necessarily incorrect, but the amount of simultaneous information can make the viewer stop interpreting and begin scanning without a goal.

The dashboard also illustrates why one interface cannot serve every audience equally well. An experienced fan may want all of these panels because hiding them would slow down analysis. A novice may need the same data introduced in stages, with definitions and a clearer indication of what changed recently.

Detailed comparison

Timing table and driver tracker

The timing table and driver tracker describe the same race from different angles. The timing table tells the viewer the order and the measured gaps. The tracker is spatial and shows where cars sit on the circuit. If the question is 'How many seconds separate these drivers?' the timing table is better. If the question is 'Is this driver about to reach traffic?' the tracker is easier to read.

For a novice, the tracker is more approachable because it resembles the circuit shown on television. Its weakness is that physical closeness does not always equal a battle for position. The timing table is harder to enter, but it is less likely to confuse track position with race position. The two displays work best together: the map shows the situation, while the table confirms the order and gap.

Timing table and tyre display

The timing table explains what is happening now, while the tyre display helps explain why the pace may change next. A small gap can mean different things depending on tyre compound and tyre age. For example, a driver on newer soft tyres may be expected to close the gap quickly, while a driver on older hard tyres may be trying to reach the end without another stop.

The problem is that the interfaces usually leave the viewer to make this connection. The timing table provides the gap; the tyre display provides the compound and age; neither plainly states the likely consequence. Experienced viewers supply that interpretation from memory. Novices need a short explanation such as 'newer tyre,' 'likely faster now,' or 'may need another stop.'

Official interfaces and the third party dashboard

The official interfaces separate race information into focused views. This keeps each screen manageable, but it also forces the viewer to switch between timing, track position, and tyre history. The third-party dashboard puts these views together. That makes comparison faster for an expert and more demanding for a beginner.

The Graphprix display is not simply worse because it is crowded. Its density matches its purpose: detailed analysis. The novice-viewer issue is the lack of a simpler entry state. A useful design could preserve the full dashboard while initially emphasizing only the running order, current battle, tyre difference, and most recent change. Additional panels could remain available when the viewer chooses to investigate further.

Which interface works best for a novice?

The driver tracker is the easiest starting point because location and movement are visible without much vocabulary. The timing table becomes more useful once the viewer understands gap references and driver codes. The tyre display is compact, but it needs the most explanation about strategy. The third-party dashboard offers the most depth and the least guidance, so it is better suited to viewers who already know what they want to examine.

Design implications

The comparison suggests that novice-friendly F1 graphics should make the reference point and the meaning of the data easier to see. Interval should be labeled as 'gap to car ahead,' and To Leader as 'gap to leader.' Tyre age should include the unit 'laps.' Compound letters should have a short explanation available without leaving the screen.

The interfaces could also identify the most important recent change. Instead of expecting the viewer to notice several moving values, the display could briefly state that a driver entered the pit lane, lost a position, set the fastest lap, or moved within one second of the car ahead. This would give the numbers a story without replacing them.

Finally, the amount of information should respond to the viewer's task. A beginner mode could show the running order, simple gaps, current tyres, and one or two race events. More detailed sector, telemetry, and stint information could remain available in an expanded view. This follows Malik's concern about one-size-fits-all dashboards while keeping the expert features that experienced fans value.

Conclusion

Each interface answers a different question well. Timing tables give precision, driver trackers provide spatial context, tyre displays support strategy reading, and third-party dashboards support detailed investigation. The difficulty for novices appears when the screen assumes that these relationships are already understood.

The clearest improvements are also fairly small: explain the reference point, write the unit, support color with text, reduce overlap, and tell the viewer what changed. These changes would not make the interfaces less useful for experienced fans. They would simply make the first few races easier to follow.

References

- Bowers, M. (updated 2026). Numbers Shouldn't Lie An Overview of Common Data Visualization Mistakes. Toptal. <https://www.toptal.com/designers/ux/data-visualization-mistakes>
- Malik, A. (2025). Top 7 Challenges in Data Visualization and How to Overcome Them. Platform 3 Solutions. <https://platform3solutions.com/blog/top-7-challenges-in-data-visualization-and-how-to-overcome-them/>
- Formula 1. F1 TV Features Upgrade More Angles More Racing More Control. <https://www.formula1.com/en/latest/article/f1-tv-features-upgrade-more-angles-more-racing-more-control.7CotTKwOMmsrEdtGFtu1Na>
- Graphprix. Pitwall Dashboard. <https://graphprix.com/>